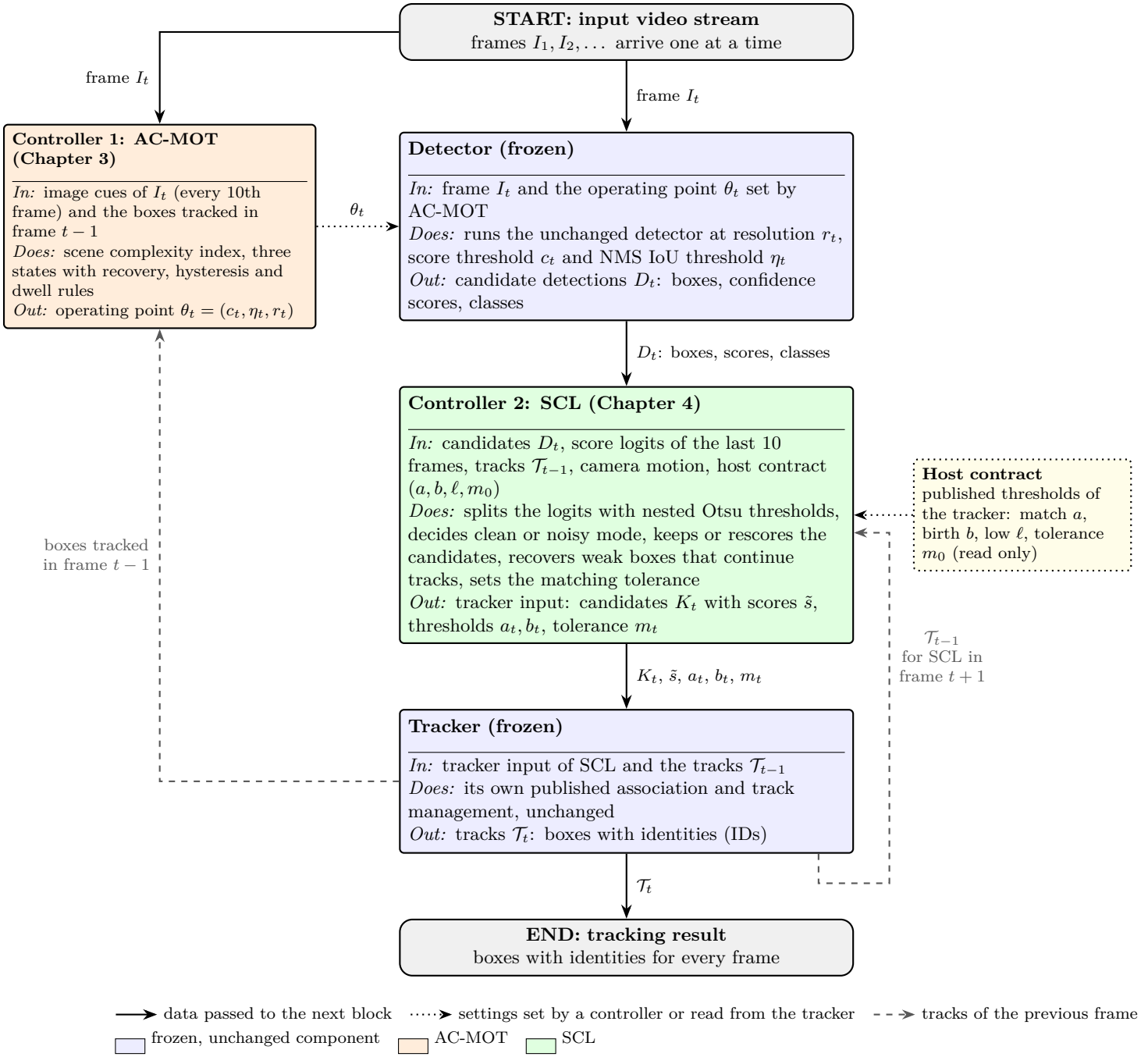

START: input video stream
frames $I_1, I_2, \ldots$ arrive one at a time
frame I_t
frame I_t
Controller 1: AC-MOT (Chapter 3)
In: image cues of I_t (every 10th frame) and the boxes tracked in frame $t-1$
Does: scene complexity index, three states with recovery, hysteresis and dwell rules
Out: operating point $\theta_t = (c_t, \eta_t, r_t)$
θ_t
Detector (frozen)
In: frame I_t and the operating point θ_t set by AC-MOT
Does: runs the unchanged detector at resolution r_t, score threshold c_t and NMS IoU threshold η_t
Out: candidate detections D_t: boxes, confidence scores, classes
D_t: boxes, scores, classes
Controller 2: SCL (Chapter 4)
In: candidates D_t, score logits of the last 10 frames, tracks $\mathcal{T}_{t-1}$, camera motion, host contract (a, b, ℓ, m_0)
Does: splits the logits with nested Otsu thresholds, decides clean or noisy mode, keeps or rescores the candidates, recovers weak boxes that continue tracks, sets the matching tolerance
Out: tracker input: candidates K_t with scores $\tilde{s}$, thresholds a_t, b_t, tolerance m_t
Host contract
published thresholds of the tracker: match a, birth b, low ℓ, tolerance m_0 (read only)
boxes tracked in frame $t-1$
$\mathcal{T}_{t-1}$ for SCL in frame $t+1$
K_t, $\tilde{s}$, a_t, b_t, m_t
Tracker (frozen)
In: tracker input of SCL and the tracks $\mathcal{T}_{t-1}$
Does: its own published association and track management, unchanged
Out: tracks $\mathcal{T}_t$: boxes with identities (IDs)
$\mathcal{T}_t$
END: tracking result
boxes with identities for every frame
data passed to the next block
settings set by a controller or read from the tracker
tracks of the previous frame
frozen, unchanged component
AC-MOT
SCL